HACK APERTUS LIEBEFELD · RESEARCH & EXPERTISE NAVIGATOR CHALLENGE

# BearMiner

## *Dig Bern gold from life-science partners.*

A Research & Expertise Navigator that turns scattered university web pages into source-backed experts, units and partners — powered by Apertus.

BUILT FOR Bern Economic Development Agency

TEMPTEAM · OCTOBER 2026



THE PROBLEM

# To find an expert today, you have to know **where** to look before you know **what** to look for.

"glucose sensors" | continuous glucose monitoring | CGM | biosensing | metabolic monitoring | wearable sensing

**3**

source ecosystems: UniBE subdomains, ARTORG, Diabetes Center Berne

**5+**

page types per institute: faculty, group, person, project, service

**0**

places a company can ask one question and get traceable answers

OUR SOLUTION

# Two sides of the same search, meeting on evidence.

RESEARCH SIDE

## What Bern can do

- Focused crawl of all three ecosystems
- Up to 500 ranked research highlights
- EN/DE medical keywords, verbatim evidence, source URL



MATCH

COMPANY SIDE

## Who Bern should call

- Reads a pitch PDF, website or text, any language
- Builds a verified medical profile
- Collaboration potential 0–100 and P1–P4 priority

HOW IT WORKS

# Public web in, source-backed answers out.

1

## Collect

Inventory hosts only; robots.txt and rate limits respected

RULES

2

## Select

Findings, projects, technologies, services, each with a quote

RULES

3

## Understand

Need → terms and EN/DE synonyms; company medical profile

APERTUS

4

## Retrieve & score

BM25 over records; five-part, recomputable match rate

RULES

5

## Explain

Keeps relevant units and writes why each one matches

APERTUS

Apertus v1.5 — language understanding

Deterministic code — crawling, verification, scoring

WHAT A RESULT LOOKS LIKE

# Every answer carries its own proof.

DIABETES CENTER BERNE · PROJECT

## The Moonwalk Project

glucose · hypoglycemia

> “After the end of the study, algorithms will be developed to build a warning system for nocturnal hypoglycemia…”

dcberne.com/en/project/the-moonwalk-project/

Real record from our live crawl test: keywords, verbatim quote, public source.

SAMPLE COMPANY · WEARABLE CGM, ZURICH

**81.1**

**Strong potential — prioritise outreach**

| | |
|---|---|
| Capability overlap | 25.8/40 |
| Partnership readiness | 17.6/20 |
| Domain fit | 14.2/15 |
| Swiss proximity | 13.5/15 |
| Signal depth | 10/10 |

Why not higher: “Swiss company but no Canton of Bern site yet.”

PURPOSEFUL USE OF AI

# Apertus where language is hard. Rules where trust matters.

## Apertus does

**Bridges vocabulary** — “glucose sensors” becomes CGM, biosensing, metabolic monitoring, in English and German

**Reads companies** — any-language pitch to a structured medical profile

**Explains matches** — a short, sourced reason per result

## Code does

**Decides the score** — same input, same number, fully offline

**Checks every claim** against the source text

**Collects and ranks** research without any model or GPU

TRACEABILITY BY DESIGN

# No invented people, roles or projects.

VERBATIM GATE

## Quote or drop

Each extracted item needs a quote found in the source (exact or ≥ 85% word match). Unsupported items are dropped and listed.

CLOSED CHOICE

## No free-form people

The judge can only select candidate IDs it was given; each result keeps its source URLs.

HONEST GAPS

## “No reliable match”

Below threshold, the system says so instead of guessing. Thin pages are marked insufficient.

PROVENANCE

## Verified location only

Bern presence earns points only from a source, JSON-LD, imprint or Wikidata — never model memory.

SAFE INPUT

## Injection defence

Web text is wrapped as untrusted data; sentences addressed to an AI are removed.

FAIR REACH

## No size bias

No institutional quota, recency bonus or mention-count boost in ranking.

TECHNICAL RIGOUR

# The score separates real fits from noise.

| SAMPLE COMPANY | MATCH RATE |
|---|---|
| Bern glucose-sensor start-up, site in Burgdorf | 88.6 |
| Wearable CGM company, Zurich | 86.8 |
| Surgical navigation, Munich | 76.1 |
| Glucose diagnostics, Japanese-language source | 52.1 |
| Medical devices, little detail | 38.2 |
| One-line “wearable sensors” pitch | 29.8 |
| Fintech — out of scope | 0.0 |

**5 → 7** live medical pages produced seven evidence-backed records with keywords

**30** labelled benchmark examples, 10 per source; build fails below 90% precision

**EN·DE·JA** sources handled — translated before matching

**Tests** for robots, redirects, budgets, evidence gates and the 500-record cap

SOVEREIGN · AFFORDABLE · EXTENSIBLE

# Runs on Bern's terms, air-gapped if needed.

## Sovereign

Apertus v1.5 on an on-premise or Swiss-cloud endpoint. SQLite and Python — no foreign closed service at runtime.

## Low cost

Crawling and scoring need no GPU. LLM answers are cached, so re-runs are instant and identical.

## Extensible

A new institution is one config entry — then re-run crawl and enrich. No pipeline rewrite.

```
# src/sources.json — add a source
{ "id": "newinst", "name": "New Institute", "url": "https://…", "institution": "…" }
```

FROM PROTOTYPE TO PILOT

# Give us one quarter and one real company inbox.



A TOOL FOR Bern Economic Development Agency

WEEKS 1–4

## Complete crawl

Full run over all three ecosystems on a Swiss-hosted Apertus endpoint.

WEEKS 5–8

## Expert review

Agency staff review highlights; feedback tunes rules and synonyms.

WEEKS 9–12

## Live outreach

Score incoming company requests and hand off P1 leads to Bern units.